

Introduction to Electrical Engineering Practice Exam

Discussion: January 12, 2026

1 Conceptual Understanding: Multiple Choice

Select exactly one answer per question.

1. The drift motion of electrons in a conductor is mainly caused by:

- A) temperature gradients
- B) applied magnetic fields
- C) an external electric field
- D) mechanical forces

Solution: C

2. For a metallic conductor, an increase in temperature usually leads to:

- A) decreasing resistance
- B) sudden insulating behavior
- C) unchanged resistance
- D) increasing resistance

Solution: D

3. Kirchhoff's current law states:

- A) The sum of voltages in a loop is zero.
- B) The sum of currents at a node is zero.
- C) The sum of resistances at a node is constant.
- D) Power is stored at a node.

Solution: B

4. An ideal voltage source is characterized by:

- A) a fixed voltage independent of current
- B) a fixed current independent of the load
- C) infinite internal resistance
- D) no influence on connected circuits

Solution: A

5. The nodal-voltage method primarily works with:

- A) mesh currents
- B) node voltages
- C) equivalent sources
- D) frequency analysis

Solution: B

6. For maximum power transfer from a source to a purely resistive load:

- A) load resistance $>$ source internal resistance
- B) load resistance $<$ source internal resistance
- C) load resistance $=$ source internal resistance
- D) load resistance equals zero

Solution: C

7. An ammeter is typically connected:

- A) in parallel with the measured quantity
- B) in series with the measured quantity
- C) with zero internal resistance
- D) only in DC circuits

Solution: B

8. A capacitor primarily stores:

- A) magnetic energy
- B) mechanical energy
- C) thermal energy
- D) electric energy

Solution: D

9. Under constant DC voltage, a capacitor behaves (in steady state) like:

- A) a short circuit
- B) an open circuit
- C) a resistor with fixed value
- D) a voltage source

Solution: B

10. Magnetic fields are mainly generated by:

- A) stationary charges
- B) moving charges or electric current
- C) temperature changes
- D) light radiation

Solution: B

11. A conducting loop is located in a magnetic field that is constant in time. Electrical induction occurs when:

- A) the loop rotates
- B) the loop remains at rest
- C) the magnetic field is homogeneous
- D) the electrical resistance of the loop increases

Solution: A

12. Why can the current through an ideal inductor not change abruptly?

- A) Because the magnetic energy of the inductor would become infinite for a sudden change in current.
- B) Because an abrupt change in current would require an infinite induced voltage.
- C) Because magnetic fields cannot be established instantaneously.
- D) Because the inductance depends on the current.

Solution: B

13. An ideal transformer can:

- A) convert DC voltage to different DC voltage
- B) convert AC voltage to AC voltage with different amplitude
- C) permanently change the frequency of AC voltage
- D) generate energy

Solution: B

14. A transient process in an electrical network is characterized by:

- A) an algebraic equation without time dependence
- B) a first-order homogeneous differential equation with an exponential solution
- C) the absence of energy storage elements

D) a time-invariant (constant) state

Solution: B

15. In which situation is the phasor representation of a current *not* applicable?

A) for a sinusoidal current with constant frequency

B) in the steady-state of a linear network

C) during an exponentially decaying transient

D) for an AC voltage with fixed angular frequency

Solution: C

16. How does the reactance of a capacitor change with increasing frequency?

A) It increases.

B) It decreases.

C) It remains constant.

D) It is unpredictable.

Solution: B

17. Reactive power occurs in an AC circuit when:

A) voltage and current are phase-shifted

B) voltage and current are exactly in phase

C) no energy is transferred

D) only DC current flows

Solution: A

18. A high-pass filter ideally allows:

A) low frequencies

B) high voltages

C) DC only

D) high frequencies

Solution: D

19. Resonance in a series resonant circuit means:

A) minimum impedance

B) maximum damping

C) no load influence

D) instantaneous decay

Solution: A

20. A practical advantage of a symmetrical three-phase system is:

A) lower frequency

B) only two conductors are required

C) generation of a rotating magnetic field in electric machines

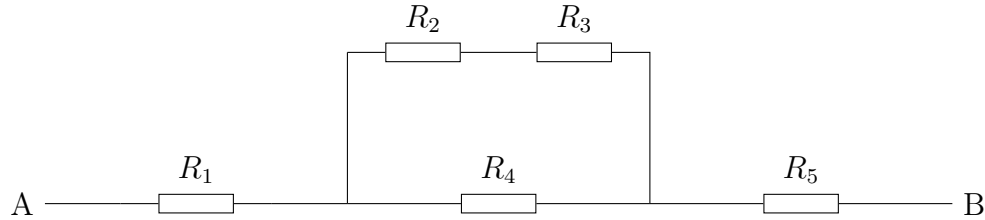
D) no balancing current possible

Solution: C

2 DC Networks

2.1 Equivalent Resistance

Given the resistor network with $R_1 = 10\ \Omega$, $R_2 = 20\ \Omega$, $R_3 = 30\ \Omega$, $R_4 = 5\ \Omega$, $R_5 = 15\ \Omega$:



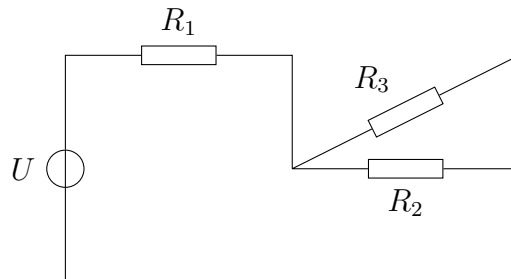
Calculate the equivalent resistance of the network between terminals A and B.

Solution:

$$\begin{aligned}
 R_{23} &= R_2 + R_3 = 20 + 30 = 50\ \Omega \\
 \frac{1}{R_{234}} &= \frac{1}{R_{23}} + \frac{1}{R_4} = \frac{1}{50} + \frac{1}{5} = \frac{11}{50} \frac{1}{\Omega} \\
 R &= R_1 + R_{234} + R_5 = 10 + \frac{50}{11} + 15 = 29.55\ \Omega
 \end{aligned}$$

2.2 Network Analysis

Given the circuit with $R_1 = 6\ \Omega$, $R_2 = 12\ \Omega$, $R_3 = 4\ \Omega$ and a voltage source of 24 V :



Determine:

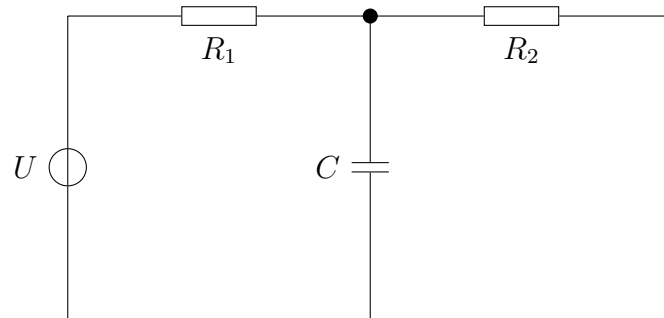
1. the total source current,
2. the branch currents through R_2 and R_3 ,
3. the power dissipated in R_3 .

Solution:

$$\begin{aligned}
 R_{23} &= \frac{12 \cdot 4}{16} = 3\ \Omega \\
 R &= 6 + 3 = 9\ \Omega \\
 I &= \frac{24}{9} = 2.67\text{ A} \\
 U_{23} &= I \cdot 3 = 8\text{ V} \\
 I_2 &= \frac{8}{12} = 0.67\text{ A}, \quad I_3 = \frac{8}{4} = 2\text{ A} \\
 P_3 &= I_3^2 R_3 = 16\text{ W}
 \end{aligned}$$

2.3 Nodal Analysis

Given the DC network with $U = 20\text{ V}$, $R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$, $C = 100\ \mu\text{F}$:



The lower conductor is chosen as reference potential (ground). The steady-state condition ($t \rightarrow \infty$) is assumed.

Determine:

1. the node voltage at the marked node,
2. the voltage across the capacitor,
3. the current through the capacitor,
4. the energy stored in the capacitor.

Solution:

$$\frac{V - U}{R_1} + \frac{V - 0}{R_2} = 0$$

$$2(V - 20) + V = 0$$

$$V = \frac{40}{3} \approx 13.33\text{ V}$$

$$U_C = V - 0 = 13.33\text{ V}$$

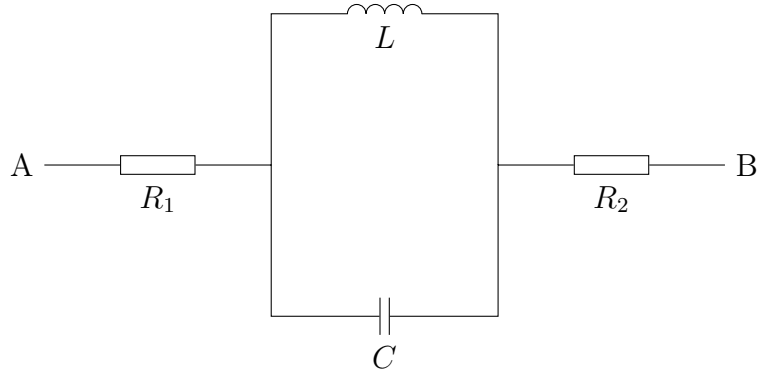
$$i_C = C \frac{du_C}{dt} = 0$$

$$W = \frac{1}{2} C U_C^2 \approx 8.9\text{ mJ}$$

3 AC Networks

3.1 Equivalent Impedance

Given the AC network with $R_1 = 10\ \Omega$, $R_2 = 20\ \Omega$, $L = 50\ \text{mH}$, $C = 100\ \mu\text{F}$, $\omega = 1000\ \text{rad/s}$:



Determine:

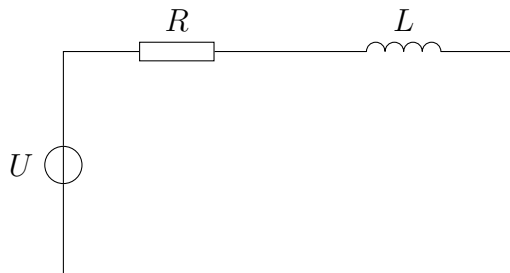
1. the complex impedances of the individual components,
2. the complex equivalent impedance between terminals A and B,
3. the magnitude and phase of the equivalent impedance.

Solution:

$$\begin{aligned}\underline{Z}_R &= R \\ \underline{Z}_L &= j\omega L = j \cdot 1000 \cdot 0.05 = j50\ \Omega \\ \underline{Z}_C &= \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 100 \cdot 10^{-6}} = -j10\ \Omega \\ \underline{Z}_{LC} &= \left(\frac{1}{\underline{Z}_L} + \frac{1}{\underline{Z}_C} \right)^{-1} = \frac{1}{j0.08} = -j12.5\ \Omega \\ \underline{Z} &= R_1 + \underline{Z}_{LC} + R_2 = (30 - j12.5)\ \Omega \\ |\underline{Z}| &= \sqrt{30^2 + 12.5^2} \approx 32.5\ \Omega \\ \varphi &= \arctan\left(\frac{-12.5}{30}\right) \approx -22.6^\circ\end{aligned}$$

3.2 Complex Network Analysis

Given the following network with a sinusoidal voltage source $U = 230\ \text{V}$, $f = 50\ \text{Hz}$ and the components $R = 20\ \Omega$, $L = 100\ \text{mH}$:



Determine:

1. the complex impedance Z ,
2. the RMS current I ,
3. the phase shift between current and voltage,
4. the power factor $\cos \varphi$,
5. the active power P ,
6. the reactive power Q .

Solution:

$$\omega = 2\pi f = 314 \text{ rad/s}$$

$$X_L = \omega L = 31.4 \Omega$$

$$\underline{Z} = (20 + j31.4) \Omega \Rightarrow |\underline{Z}| = 37.2 \Omega$$

$$I = \frac{230}{37.2} = 6.18 \text{ A}$$

$$\varphi = \arctan\left(\frac{31.4}{20}\right) = 57.5^\circ$$

$$\cos \varphi = 0.54$$

$$P = UI \cos \varphi = 766 \text{ W}$$

$$Q = UI \sin \varphi = 1180 \text{ VAR}$$

Formula Sheet

Capacitor

$$i_C(t) = C \frac{du_C(t)}{dt}$$

$$W = \frac{1}{2}CU^2$$

AC

$$\omega = 2\pi f$$

$$\underline{Z}_L = j\omega L$$

$$\underline{Z}_C = \frac{1}{j\omega C}$$

$$\underline{Z} = R + jX$$

$$P = UI \cos \varphi$$

$$Q = UI \sin \varphi$$